II BE (Civil Engineering)
Class Test III November- 2023

SUB CODE: 3VLRG1 **SUB. NAME: STRUCTURAL MECHNICS**
Duration: 70 Min **Maximum Marks: 20**

Note: Attempt any two questions. All questions carry equal marks.

Q.1. Analyse the following continuous beam as shown in figure-1by "Three Moment Equation" and draw BMD.

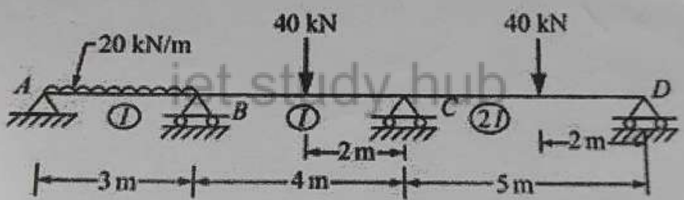

FIG – 1

Q.2. State the assumptions made in Euler's theory for column. Derive the expression for Euler's crippling load for a long column when both ends are fixed.

Q.3. A hollow cylinder cast iron column is 4 m long with both ends fixed. Determine the minimum diameter of column if it has to carry a safe load of 250 kN with a factor of safety of 5. Take internal diameter as 0.8 times the external diameter. Take $\sigma_C = 550\ N/mm^2$ and a $= 1/1600$ in Rankine's formula.